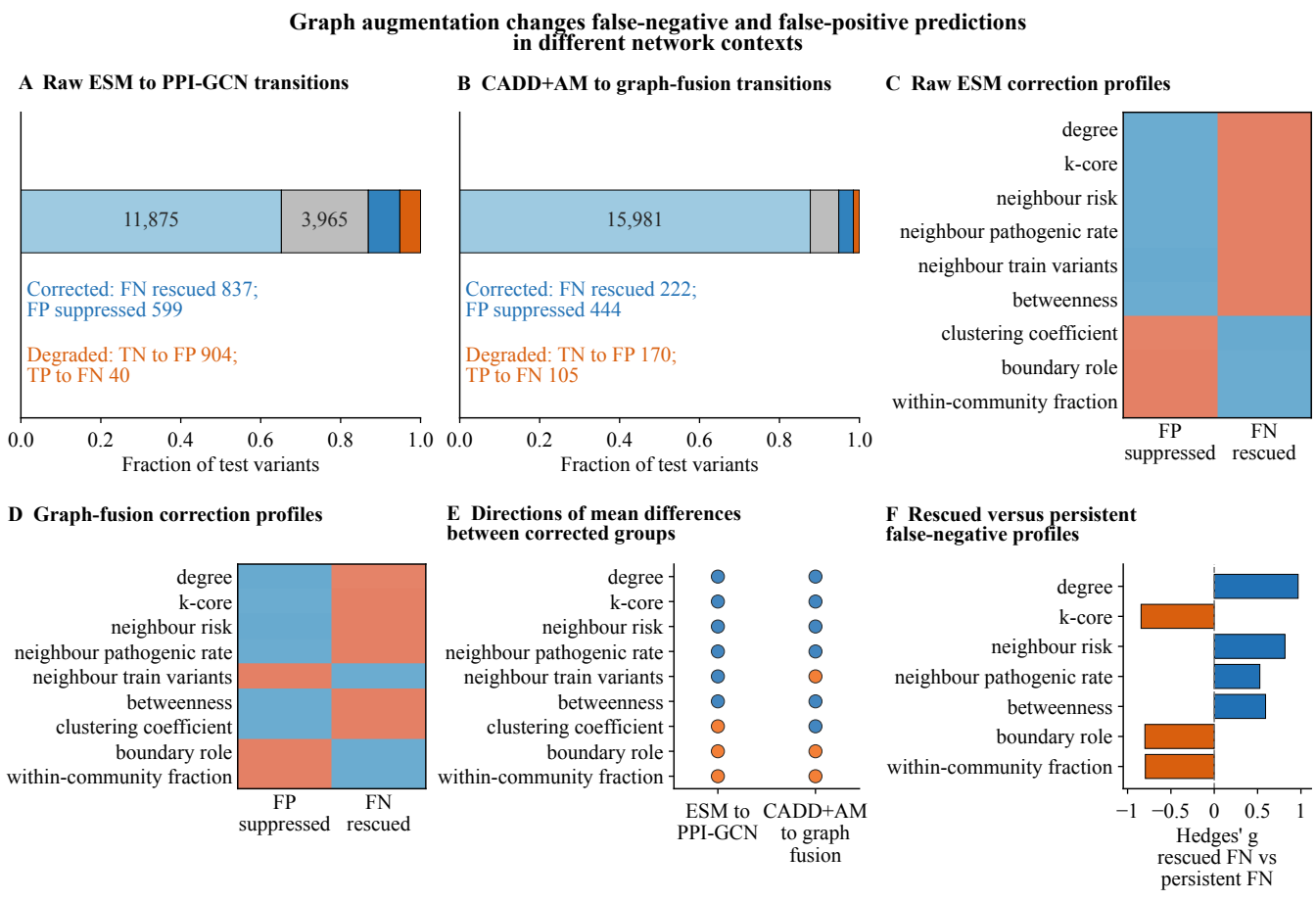

Graph augmentation changes false-negative and false-positive predictions in different network contexts
A Raw ESM to PPI-GCN transitions
11,875
3,965
Corrected: FN rescued 837; FP suppressed 599
Degraded: TN to FP 904; TP to FN 40
0.0 0.2 0.4 0.6 0.8 1.0
Fraction of test variants
B CADD+AM to graph-fusion transitions
15,981
Corrected: FN rescued 222; FP suppressed 444
Degraded: TN to FP 170; TP to FN 105
0.0 0.2 0.4 0.6 0.8 1.0
Fraction of test variants
C Raw ESM correction profiles
degree
k-core
neighbour risk
neighbour pathogenic rate
neighbour train variants
betweenness
clustering coefficient
boundary role
within-community fraction
FP suppressed
FN rescued
D Graph-fusion correction profiles
degree
k-core
neighbour risk
neighbour pathogenic rate
neighbour train variants
betweenness
clustering coefficient
boundary role
within-community fraction
FP suppressed
FN rescued
E Directions of mean differences between corrected groups
degree
k-core
neighbour risk
neighbour pathogenic rate
neighbour train variants
betweenness
clustering coefficient
boundary role
within-community fraction
ESM to PPI-GCN
CADD+AM to graph fusion
F Rescued versus persistent false-negative profiles
degree
k-core
neighbour risk
neighbour pathogenic rate
betweenness
boundary role
within-community fraction
−1 −0.5 0 0.5 1
Hedges' g
rescued FN vs persistent FN